

PAPER_1146: Heterotic String Theory and SCm Gauge Sector

Star Magic UQFF Framework

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Abstract

We formulate heterotic string theory within the SCm vacuum framework, where the gauge group ($E8 \times E8$ or $SO(32)$) emerges from the chirality of the SCm phonon modes. The left-moving sector (26D bosonic = gauge sector) is driven by the $S_{26}^{(3)}$ Ramanujan amplification, while the right-moving sector (10D superstring = gravity sector) is driven by the Φ_{res} resonance phase. The chirality splitting is mediated by the negative-time gate $\cos(\pi t_n)$, which acts as a chiral projector separating left-movers from right-movers.

1. Heterotic String Construction

The heterotic string has: - **Left-movers:** 26D bosonic string ($D_L = 26 = \text{all VDS rungs}$) - **Right-movers:** 10D superstring ($D_R = 10 = \text{SCm 4D} + 6 \text{ CY dimensions}$) - **Gauge group:** from the 16 extra left-moving dimensions ($26 - 10 = 16$)

The 16 extra left-moving dimensions compactify on a self-dual even lattice, giving $E8 \times E8$ or $SO(32)$.

2. SCm Chirality Split

The $\cos(\pi t_n)$ gate acts as a chiral projector:

$$P_L = \frac{1 + \cos(\pi t_n)}{2}, \quad P_R = \frac{1 - \cos(\pi t_n)}{2}$$

At $t_n = 0$ (forward time): $P_L = 1, P_R = 0$ — pure left-mover vacuum. At $t_n = 1$ (half negative-time cycle): $P_L = 1/2, P_R = 1/2$ — balanced. At $t_n = -1$ (full negative-time): $P_L = 1, P_R = 0$ — also pure left-mover.

The chirality is maximum at $t_n = 1/2$: $P_L = P_R = 1/2$.

3. $E8 \times E8$ from VDS Rungs

The 16 gauge dimensions split into two sets of 8, corresponding to VDS rungs 11-18 and 19-26:

$$E8_1 \leftrightarrow \{[SSq]^{11}, \dots, [SSq]^{18}\}$$

$$E8_2 \leftrightarrow \{[SSq]^{19}, \dots, [SSq]^{26}\}$$

The E8 root lattice radius in SCm:

$$R_{\text{E8}} = l_s \cdot [SSq]^{18/2} = l_s \cdot (0.57)^9 = l_s \times 2.29 \times 10^{-7}$$

4. Gauge Field Action

The 10D heterotic effective action:

$$S_{\text{het}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x e^{-2\phi} \sqrt{-g} \left(R + 4|\partial\phi|^2 - \frac{1}{2}|H_3|^2 - \frac{\alpha'}{4}\text{tr}|F_2|^2 \right)$$

where F_2 is the E8×E8 gauge field strength and $\alpha' = l_s^2 = 1/T$.

The SCm gauge coupling:

$$g_{\text{YM}}^2 = \frac{\kappa_{10}^2}{l_s^8} \cdot e^{2\phi} = g_s^2 \cdot T^4 = (0.504)^2 \times (8.66 \times 10^{-11})^4$$

5. Anomaly Cancellation via SCm

The Green-Schwarz anomaly cancellation in Type I SO(32) / heterotic requires:

$$H_3 = dB_2 - \frac{\alpha'}{4} (\omega_{\text{YM}} - \omega_{\text{grav}})$$

In SCm: H_3 is the NS-NS phonon mode (as in PAPER_1144) and ω_{YM} is the Chern-Simons 3-form of the E8×E8 SCm gauge field.

6. Chirality and Fermion Masses

The standard model chirality (left-handed weak interactions) emerges from the heterotic $\cos(\pi t_n)$ chiral projector. The fermion mass hierarchy:

$$m_f \propto |\cos(\pi t_{n,f})| \cdot \beta_i \cdot \Phi_{\text{res}} \cdot v_{\text{EW}}$$

where $t_{n,f}$ is the fermion's characteristic negative-time phase.

7. Calibrated Constants

$$T = 8.66 \times 10^{-11} \text{ N}, \quad [SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$g_s = 0.504, \quad \Phi_{\text{res}} = 0.84, \quad \beta_i = 0.6, \quad f_{\text{THz}} = 1.25 \text{ THz}$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. Gross, D.J. et al. (1985). Heterotic string theory (I). *Nucl. Phys. B* **256**, 253.
2. Green, M.B. & Schwarz, J.H. (1984). Anomaly cancellations in supersymmetric D=10 gauge theory. *Phys. Lett. B* **149**, 117.
3. Type IIA SCm: PAPER_1145; Calabi-Yau: PAPER_1147; `scm_vacuum_manifold.py`